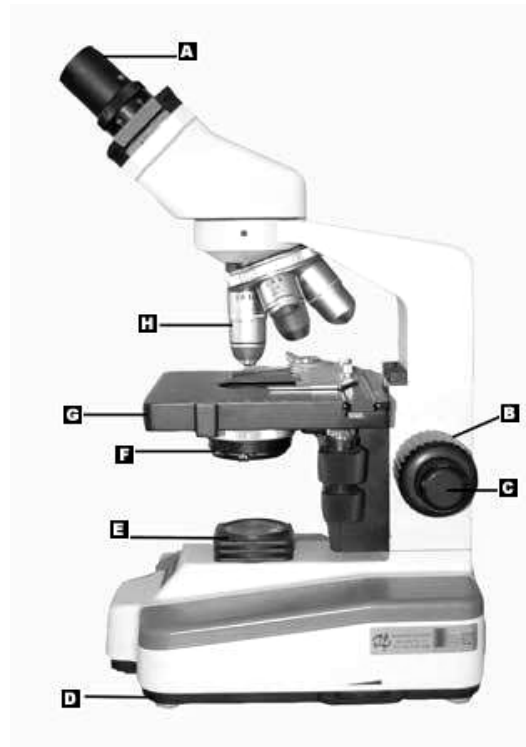


**YEAR 8 SCIENCE
BIOLOGICAL SCIENCE
REVISION - MICROSCOPES + CELLS**

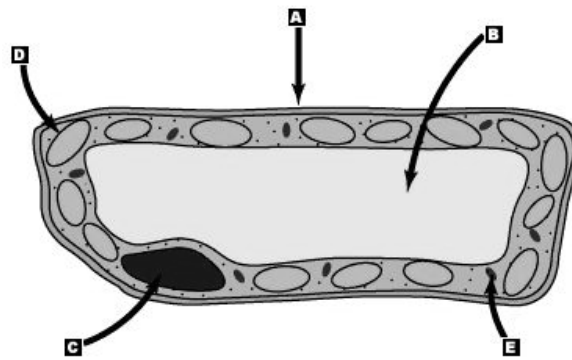
1. (a) Label the following diagram of a microscope.



- (b) It is suggested that you look at the slide from the side when switching from low to high power. Why is this?
- (c) If the eyepiece is X10 and the objective lens is X4, what is the magnification of the slide?
- (d) If you look into the microscope and observe the slide in focus, what happens to the image if you move the slide to the *right*?
- (e) The microscopes used at Holy Cross have three objective lenses that can be adjusted to give low, medium and high power viewing of slides.

When is it best to use the *fine focus knob* when viewing a slide? Explain why.

2. (a) Label the following diagram of a cell.



- (b) Is it a plant or animal cell? How do you know?

- (c) Name the parts of the cell that do the following functions.

Part of Cell	Function
	Controls the activities of the cell.
	Storage area for substances needed by the cell.
	Controls substances entering and leaving cells.
	Contains chlorophyll and is the site for photosynthesis.
	Only found in plants – gives the cell strength and helps plants stay upright.
	Produces energy for the cell.
	Jelly-like liquid that fills up most of the space in a cell.

3. List three ways in which a plant cell is different to an animal cell.

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4. Unscramble the words using the clues provided.

- (a) Control centre of the cell. SEUNCLU _____
- (b) Surrounds the cell. ERAMMBNE _____
- (c) Contains cell sap. OCVAUEL _____
- (d) Part of the cell between the cell membrane and the nucleus. CATOPLMYS _____
- (e) Building blocks of all living things. LELSC _____
- (f) Living things . ASMOGNIRS _____

5. Copy and complete the table below.

Cell feature	Plant cells	Animal cells	Fungal cells
Cell wall	✓	✗	
Cytoplasm			
Cell membrane			
Chloroplast			
Nucleus			
Large vacuole			